Name - Shreyas Shashwat
Roll no. - 20EC01007
Section - 4 Group - 1

FRESNEL'S BIPRISM

Date of experiment: **Partner's name:**

Date of Lab. report submission:

Aim of the experiment

To determine the wave length of the given source using Fresnel's biprism.

Apparatus Required

(i) Fresnel's biprism set up
(ii) Optical bench
(iii) LASER source
(iv) White screen with stand
(v) Two lenses with focal length 20 mm & 300 mm
(vi) Measuring tape

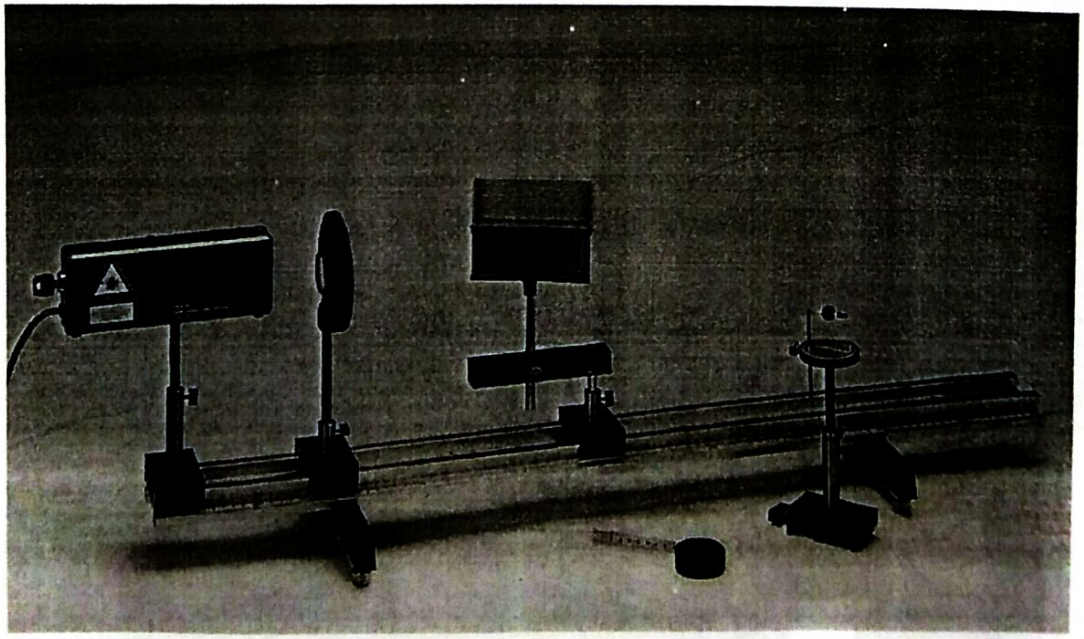

Fig-1: Experimental setup

Theory

A biprism consists of two acute angled prisms placed base to base. The experimental set up and the schematic diagram of a typical biprism is shown in Fig-1 and Fig-2 respectively. A biprism acts as a single prism with an obtuse angle 179^0 and two equal acute angles of $30'$ each. When light falls on the lower part of the biprism it bends upward and appears as if it has come from the virtual source B. Similarly when light fall on the upper part of the biprism, it bends down ward and appears as if it has come from the virtual source A. Thus A and B act as two coherent virtual sources and we get interference fringes of equal width at the screen between E and F.

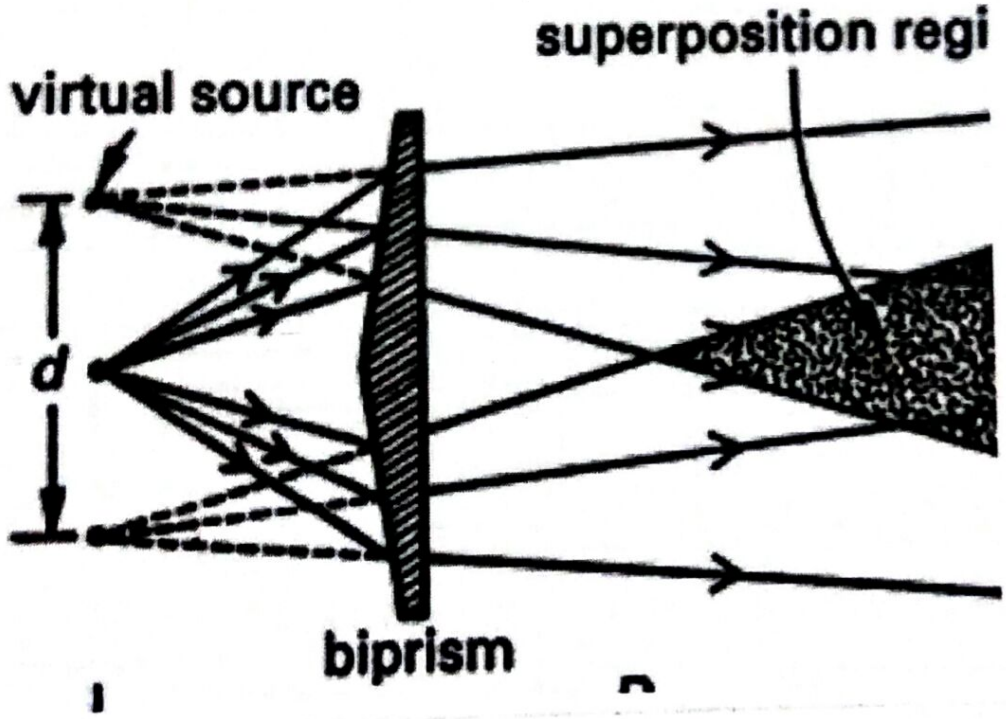

Fig-2: Schematic diagram

From the theory of interference of light we have, $\beta = \frac{\lambda D}{d}$

Where β is the width of the bright or the dark fringe

λ is the wave length of the light used

D is the distance of the screen from the source

d is the distance between the two virtual sources

To measure d, a lens of focal length 300 mm is placed between the biprism and the screen such that the real images of the two virtual sources are formed on the screen and the distance between these real images 'b' is measured.

From basic optics we know that $\frac{b}{d} = \frac{v_1}{u_1}$ or $d = \frac{bu_1}{v_1}$

where u_1 is the distance of the virtual source from the lens and v_1 is the distance of the screen from the lens. The value of u_1 is found out by applying the lens formula

$$u_1 = \frac{Fv_1}{v_1 - F}$$

So we obtain $$d = \frac{bF}{(v_1 - F)}.$$

Thus obtaining the values of d, β and D the value of λ can be determined by the formula $\lambda = \frac{\beta d}{u_1 + v_1}$.

Procedure

(i) Place the source and the lenses of focal length 20mm at appropriate positions on the optical bench.

(ii) Place the biprism on the prism table and adjust it such that the light from the source falls exactly on the centre of the biprism. Place the screen with a ruler paper fixed on it at an approximate distance of 2 m from the source.

(iii) The screen should be perpendicular to the optical bench. Adjust the prism by rotating it slightly so that a number of parallel fringes of equal width are formed on the screen.

(iv) Measure the width of 10 consecutive bright fringes.

(v) Place the lens of focal length 300 mm on the optical bench. Adjust the position of the lens such that the two real images of the virtual sources are formed on the screen.

(vi) Measure the distance b between the two images of the two virtual sources with the help of a scale.

(vii) Repeat these observations for 4 to 5 times for different positions of the screen.

Observations

Table-1: Position of various optical components

Source Position (cm)	Lens L_1 (cm)	Biprism (cm)	Lens L_2 (cm)	Screen (cm)
0.2	19.8	37.2	42.8	200
0.2	19.8	37.2	42.8	190
0.2	19.8	37.2	42.8	210

Table-2: Measurement of fringe width

Width of 10 fringes (cm)	Fringe width (β) (cm)	Mean fringe width (β) (cm)
0.6	0.06	
0.6	0.06	0.06
0.6	0.06	

Table-3: Measurement of the separation of the virtual sources

Object to lens distance (u_1) (cm)	Image to lens distance (v_1) (cm)	Separation between the real images (b) (cm)	Separation between the virtual images $d = \frac{bF}{(v_1 - F)}$ (cm)
37.07	157.2	0.6	0.1415
38.18	140.8	0.7	0.1895
37.84	144.8	0.7	0.1829

Conclusions

Conclusion should incorporate

(v) Interesting results you found in the experiment.

(vi) Lacunas /difficulties that you encountered.

(vii) Practical applications/implications of the experiment.

(viii) Any other issues.

Precautions

(i) One should not look directly into LASER light because it may damage the eye.

(ii) The optical bench should be levelled properly.

(iii) The refracting edge should be normal to the direction of the light.

(iv) The width of the fringe should be accurately measured.

(v) As many fringes as possible should be taken to find out the mean fringe width β.

References

1. Fundamentals of optics by F. Jenkins and H. White.
2. Optics by A. Ghatak.
3. Introduction to Optics by E. Hecht
4. Feynman Lectures.

Considerations

(i) What is interference of light and what are the different ways in which light interferes?

(ii) What is wave theory of light?

(iii) Is the Fresnel's biprism experiment better than Young's double slit experiment? Justify your answer.

(iv) What are coherent sources and how coherent sources are realized in biprism experiment?

(v) What will happen if we increase the focal length of the lens?

Calculations

$$u_1 = \left(\frac{fv_1}{v_1 - f}\right) = \frac{30 \times 157.2}{157.2 - 30} = \frac{4716}{127.2} = 37.07 \text{ cm}$$

$$d = \frac{bf}{v_1 - f} = \frac{0.6 \times 30}{157.2 - 30} = \frac{18}{127.2} = 0.1415 \text{ cm}$$

$$\lambda_1 = \frac{\beta d}{u_1 + v_1} = \frac{0.06 \times 0.1415}{(37.07 + 157.2)}$$

$$= \frac{0.00849}{194.27}$$

$$= 4.37 \times 10^{-5} \text{ cm}$$

$$= 437 \text{ nm}$$

$$u_1 = \frac{f v_1}{v_1 - f} = \frac{30 \times 140.8}{140.8 - 30} = \frac{4224}{110.8} = 38.12 \text{ cm}$$

$$d = \frac{bf}{v_1 - f} = \frac{0.7 \times 30}{140.8 - 30} = \frac{21}{110.8} = 0.1895 \text{ cm}$$

$$\lambda_2 = \frac{\beta d}{u_1 + v_1} = \frac{0.06 \times 0.1895}{38.12 + 140.8} = \frac{0.01137}{178.92}$$

$$= 6.3548 \times 10^{-5} \text{ cm}$$

$$= 635 \text{ nm}$$

$$u_1 = \frac{f v_1}{v_1 - f} = \frac{30 \times 144.8}{144.8 - 30} = \frac{4344}{114.8} = 37.84 \text{ cm}$$

$$d = \frac{bf}{v_1 - f} = \frac{0.7 \times 30}{144.8 - 30} = \frac{21}{114.8} = 0.1829 \text{ cm}$$

$$\lambda_3 = \frac{0.06 \times 0.1829}{37.84 + 144.8} = \frac{0.010974}{182.64} = 6.0085 \times 10^{-5} \text{ cm}$$

$$= 601 \text{ nm}$$

$$\bar{\lambda} = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3} = \frac{437 + 601 + 635}{3} = 557.67 \text{ nm}$$

$$d_1 = \bar{\lambda} - \lambda_1 = (557.67 - 437) \text{ nm} = 120.67 \text{ nm}$$

$$d_2 = \bar{\lambda} - \lambda_2 = (557.67 - 635) \text{ nm} = -77.33 \text{ nm}$$

$$d_3 = \bar{\lambda} - \lambda_3 = (557.67 - 601) \text{ nm} = -43.33 \text{ nm}$$

$$\sigma^2 = \frac{\sum d_i^2}{n} = \frac{(120.67)^2 + (77.33)^2 + (43.33)^2}{3}$$

$$= (7472.8889)$$

$$\sigma_n = \sqrt{\frac{\sigma^2}{n-1}} = 61.13 \text{ nm}$$

$$\therefore \lambda = (557.67 \pm 61.13 \text{ nm})$$

Conclusion :

1. This experiment can be used to find the wavelength of monochromatic source.

2. The wavelength of the source so obtained is (557.67 ± 61.13) nm.

3. This experiment can be used to obtain a virtual coherent source.

4. This experiment can also be used to determine the thickness of a thin transparent sheet.